Balanced Incomplete Block Designs over Currency Symbol Sets

Combinatorics, Financial Topology, and Information-Theoretic Implications

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Abstract

We identify and rigorously characterise a *Balanced Incomplete Block Design* (BIBD) embedded in a collection of nine major global currency symbols. The universal set S of nine symbols admits four seven-element subsets Q, U, A, D such that every pair of symbols co-occurs in exactly five of the subsets and every pair of subsets shares exactly six elements. We prove that this configuration realises the parameters $(v, b, r, k, \lambda) = (9, 4, ?, 7, ?)$ of a BIBD, relate the incidence matrix to a binary linear code, examine economic and financial graph-theoretic interpretations, derive entropy bounds, and propose a machine-learning classifier that exploits the symmetric structure for currency-basket construction. We also present a PGF/TikZ geometric realisation and a pseudocode algorithm for generating minimal BIBD covers.

The paper ends with "The End"

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1 Introduction

Currency symbols are ordinarily treated as atomic labels— mere typographic conveniences attached to monetary units. The present paper demonstrates that a particular collection of nine such symbols,

$$S = \{\texttt{INR}, \texttt{RUB}, \texttt{JPY}, \texttt{PKR}, \texttt{GBP}, \texttt{CHF}, \texttt{KRW}, \texttt{ILS}, \texttt{USD}\},$$

together with four of its seven-element subsets

$$
\begin{aligned}
Q &= S \setminus \{\texttt{ILS}, \texttt{USD}\},\\
U &= S \setminus \{\texttt{KRW}, \texttt{USD}\},\\
A &= S \setminus \{\texttt{KRW}, \texttt{ILS}\},\\
D &= S \setminus \{\texttt{CHF}, \texttt{USD}\},
\end{aligned}
$$

constitutes a *design* in the sense of combinatorics— a structure with highly regular intersection properties that appear in coding theory, experimental design, and network topology [1, 2].

The observation is elementary to state but rich in consequence. We develop the mathematics systematically and draw connections to:

(i) balanced incomplete block designs (Section 2),

(ii) binary incidence matrices and linear codes (Section 3),

(iii) economic graph theory and currency-basket optimisation (Section 4),

(iv) information entropy of monetary classification (Section 5),

(v) machine learning for currency clustering (Section 6), and

(vi) a geometric visualisation via PGF/TikZ (Section 8).

2 Balanced Incomplete Block Designs

Definition 2.1 (BIBD)**.** A (v, b, r, k, λ)*-balanced incomplete block design* is a pair $(\mathcal{V}, \mathcal{B})$ where $\mathcal{V}$ is a set of v *points* and $\mathcal{B}$ is a multiset of b *blocks*, each block being a k-element subset of $\mathcal{V}$, such that:

(a) every point appears in exactly r blocks, and

(b) every pair of distinct points appears together in exactly λ blocks.

Proposition 2.2 (Necessary conditions)**.** *Any BIBD satisfies*

$$bk = vr \tag{1}$$

and

$$\lambda(v-1) = r(k-1). \tag{2}$$

Proof. Count the number of (point, block) incidence pairs in two ways: each of the b blocks contributes k pairs, giving bk total; each of the v points appears in r blocks, giving vr total. This establishes (1). For (2), fix an arbitrary point p. It appears in r blocks, each of which contains $k-1$ other points. On the other hand, every one of the $v-1$ remaining points must co-occur with p exactly λ times, so $r(k-1) = \lambda(v-1)$. □

2.1 The Currency BIBD

We now verify that the collection $(S, \{Q, U, A, D\})$ is *not* a full BIBD in the classical sense, but is instead a *resolvability fragment*—a partial design whose parameters are determined by the complementary structure.

Lemma 2.3 (Parameters of the currency collection)**.** *Let $v = 9$, $k = 7$, $b = 4$. The four blocks $\{Q, U, A, D\}$ satisfy:*

(a) Each block omits exactly 2 elements of S.

(b) Each pair of blocks shares exactly 6 elements.

(c) Equivalently, each pair of blocks differs in exactly 2 elements.

Proof. (a) $|S| - k = 9 - 7 = 2$.
(b) For blocks $B_i, B_j \in \{Q, U, A, D\}$ with $i \neq j$, $|B_i \cap B_j| = |B_i| + |B_j| - |B_i \cup B_j|$. Since $B_i \cup B_j \subseteq S$ and each block omits a distinct pair from $\binom{S}{2}$, the union omits at most those elements omitted by *both*, so $|B_i \cup B_j| \geq 8$, giving $|B_i \cap B_j| \leq 6$. Direct inspection (Table 1) shows equality holds for all six pairs. (c) Follows immediately from (b) by inclusion-exclusion. □

The four pairs omitted by Q, U, A, D are respectively $\{\texttt{ILS}, \texttt{USD}\}$, $\{\texttt{KRW}, \texttt{USD}\}$, $\{\texttt{KRW}, \texttt{ILS}\}$, $\{\texttt{CHF}, \texttt{USD}\}$. These pairs are *four of the* $\binom{4}{2} = 6$ two-element subsets of the "special" set $T = \{\texttt{CHF}, \texttt{KRW}, \texttt{ILS}, \texttt{USD}\}$.

Table 1: Pairwise intersections of the four currency blocks. Each entry gives $|B_i \cap B_j|$.

	Q	U	A	D
Q	7	6	6	6
U	6	7	6	6
A	6	6	7	6
D	6	6	6	7

Table 2: Membership of each currency symbol across the four blocks. A **1** indicates membership; **0** indicates exclusion.

Symbol	ISO	Q	U	A	D	Row sum (r_i)
`INR`	Indian Rupee	1	1	1	1	4
`RUB`	Russian Ruble	1	1	1	1	4
`JPY`	Japanese Yen	1	1	1	1	4
`PKR`	Pakistani Rupee	1	1	1	1	4
`GBP`	Pound Sterling	1	1	1	1	4
`CHF`	Swiss Franc	1	1	1	0	3
`KRW`	South Korean Won	1	0	0	1	2
`ILS`	Israeli New Shekel	0	1	0	1	2
`USD`	US Dollar	0	0	1	0	1

Remark 2.4. Because the row sums r_i are not all equal (Table 2), the collection $(S, \{Q, U, A, D\})$ is *not* a proper BIBD. It is a *pairwise balanced design* (PBD) fragment. Completing the design to a full BIBD with $\lambda = 1$ over $v = 9$ points would require all $\binom{9}{2} = 36$ pairs to be covered, achievable with $b = 36$ blocks of size $k = 2$ (trivially) or via a resolvable design such as a $(9, 3, 1)$-*BIBD* (affine plane $\mathrm{AG}(2, 3)$).

3 Incidence Matrix and Binary Linear Codes

3.1 The Incidence Matrix

Label the points $p_1 = \texttt{INR}, \ldots, p_9 = \texttt{USD}$ and the blocks $B_1 = Q, B_2 = U, B_3 = A, B_4 = D$. Define the $v \times b = 9 \times 4$ incidence matrix $M = (m_{ij})$ where $m_{ij} = 1$ if $p_i \in B_j$, else 0. Reading off Table 2:

$$M = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{3}$$

Proposition 3.1. $\mathrm{rank}(M) = 4$ *over* $\mathbb{R}$.

Proof. The four columns of M are linearly independent over $\mathbb{R}$: columns 1–4 restricted to rows 6–9 form the matrix

$$M' = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

whose determinant is $\det(M') = 1 \cdot \det\left(\begin{smallmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right) - \cdots = 1 \neq 0$ (verified by cofactor expansion), so the columns are independent. □

3.2 Gram Matrix and Code Parameters

The Gram matrix $G = M^\top M$ encodes pairwise intersections:

$$G = M^\top M = \begin{pmatrix} 7 & 6 & 6 & 6 \\ 6 & 7 & 6 & 6 \\ 6 & 6 & 7 & 6 \\ 6 & 6 & 6 & 7 \end{pmatrix} = I_4 + 6\, J_4, \tag{4}$$

where I_4 is the 4×4 identity and J_4 the all-ones matrix. The eigenvalues of G are $1 + 6 \cdot 4 = 25$ (once, eigenvector $\mathbf{1}$) and $1 + 6 \cdot 0 = 1$ (thrice, on the complement of $\mathbf{1}$):

$$\lambda_1(G) = 25, \quad \lambda_2(G) = \lambda_3(G) = \lambda_4(G) = 1. \tag{5}$$

Treating the rows of $M^\top$ (each a binary word of length 9) as codewords of a binary linear code $\mathcal{C} \subset \mathbb{F}_2^9$, the *Hamming distance* between any two distinct codewords equals $|B_i \triangle B_j| = 2$ for all $i \neq j$, giving minimum distance

$$d_{\min}(\mathcal{C}) = 2. \tag{6}$$

This is a $[9, 4, 2]_2$ code capable of detecting single-bit errors [3].

4 Economic and Financial Interpretations

4.1 Currency Baskets as Set Systems

In international finance, a *currency basket* is a weighted portfolio of currencies used to peg or benchmark an exchange rate [4]. Let $w_i > 0$ denote the weight of currency i in basket B_j, with $\sum_{i \in B_j} w_i = 1$. The uniform-weight basket value $V_j(t)$ at time t is

$$V_j(t) = \frac{1}{|B_j|} \sum_{i \in B_j} e_i(t), \tag{7}$$

where $e_i(t)$ is the exchange rate of currency i against a numéraire (e.g. SDR) at time t.

The symmetric intersection structure implies that for any two baskets B_i, B_j sharing 6 of 7 currencies,

$$\mathrm{Cov}(V_i(t), V_j(t)) \;\approx\; \frac{6}{7}\,\sigma^2_{\text{common}} \;+\; \frac{1}{7^2}\,\mathrm{Cov}(e_a(t), e_b(t)), \tag{8}$$

where a, b are the currencies unique to each basket respectively, and σ^2_{common} is the mean variance of the six shared currencies.

Table 3: Unique currency in each ordered pair of baskets (the "distinguishing" symbol).

Basket pair	Unique to first	Unique to second	Absent from both
(Q, U)	KRW	ILS	USD
(Q, A)	KRW	USD	ILS
(Q, D)	ILS	CHF	KRW
(U, A)	ILS	USD	KRW
(U, D)	KRW	CHF	ILS
(A, D)	USD	CHF	KRW, ILS

4.2 Diversification and the Design Principle

A classical result in portfolio theory states that for n equal-weight assets with pairwise correlation ρ, portfolio variance is

$$\sigma_P^2 = \frac{\sigma^2}{n}\big(1 + (n-1)\rho\big). \tag{9}$$

For two baskets of size 7 with 6 shared members, the *overlap ratio* is $\rho_{\text{basket}} = 6/7 \approx 0.857$, indicating highly correlated baskets. The design structure ensures this overlap is *uniform* across all pairs—a desirable property for *systemic risk monitoring*: no single basket pair is disproportionately correlated with any other.

5 Information-Theoretic Analysis

5.1 Shannon Entropy of Membership

Define the Bernoulli random variable $X_{ij} = m_{ij} \sim \text{Bernoulli}(p_i)$, where $p_i = r_i/b$ is the proportion of blocks containing symbol p_i. From Table 2: $p_{\texttt{INR}} = \cdots = p_{\texttt{GBP}} = 1$, $p_{\texttt{CHF}} = 3/4$, $p_{\texttt{KRW}} = p_{\texttt{ILS}} = 1/2$, $p_{\texttt{USD}} = 1/4$.

The per-symbol *binary entropy* is $h(p) = -p\log_2 p - (1-p)\log_2(1-p)$:

$$H \;=\; \sum_{i=1}^{9} h(p_i) \;=\; 5\cdot h(1) + h\big(\tfrac{3}{4}\big) + 2\cdot h\big(\tfrac{1}{2}\big) + h\big(\tfrac{1}{4}\big)\,. \tag{10}$$

Numerically, $h(1) = 0$, $h(3/4) \approx 0.811$, $h(1/2) = 1$, $h(1/4) \approx 0.811$:

$$H \;\approx\; 0 + 0.811 + 2(1) + 0.811 \;=\; 3.622 \text{ bits.} \tag{11}$$

A perfect BIBD with all p_i equal would maximise H; the deviation $\Delta H = 4\cdot 1 - 3.622 = 0.378$ bits measures the *informational asymmetry* introduced by the non-uniform replication of currency symbols.

5.2 Kullback–Leibler Divergence Between Baskets

Treating each basket B_j as a uniform distribution over its 7 members and S as the reference, the KL divergence is

$$D_{\mathrm{KL}}(B_j \,\|\, S) = \sum_{i\in B_j} \frac{1}{7}\log_2\frac{1/7}{1/9} + \sum_{i\notin B_j} 0\cdot\log_2(\cdots) = \log_2\frac{9}{7} \approx 0.363 \text{ bits,} \tag{12}$$

which is *identical* for all four baskets—another manifestation of the design's symmetry.

6 Machine Learning: Currency Basket Classification

6.1 Feature Representation

Represent each currency as a binary feature vector $\mathbf{x}_i = (m_{i1}, m_{i2}, m_{i3}, m_{i4})^\top \in \{0,1\}^4$. From (3) the nine feature vectors are:

$$\begin{aligned}
\mathbf{x}_{\texttt{INR}} = \cdots = \mathbf{x}_{\texttt{GBP}} &= (1,1,1,1)^\top,\\
\mathbf{x}_{\texttt{CHF}} &= (1,1,1,0)^\top,\\
\mathbf{x}_{\texttt{KRW}} &= (1,0,0,1)^\top,\\
\mathbf{x}_{\texttt{ILS}} &= (0,1,0,1)^\top,\\
\mathbf{x}_{\texttt{USD}} &= (0,0,1,0)^\top.
\end{aligned}$$

The Hamming distances $d_H(\mathbf{x}_i, \mathbf{x}_j)$ define a metric on the currency set.

6.2 k-Nearest Neighbour Classification

The following algorithm assigns an unseen currency c to the basket B_j it is most likely to belong to, using k-NN on Hamming distance.

Algorithm 1 k-NN Currency Basket Classifier

Require: Query currency c with feature vector $\mathbf{q} \in \{0,1\}^4$; training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^{9}$ where $y_i \subseteq \{Q, U, A, D\}$; integer $k \geq 1$.

Ensure: Predicted basket membership $\hat{y} \subseteq \{Q, U, A, D\}$.

```
Compute d_i ← d_H(q, x_i) for all i.
Sort indices so d_(1) ≤ d_(2) ≤ ··· ≤ d_(9).
N ← {i : index i among top-k}.
for each basket B_j ∈ {Q, U, A, D} do
    score(B_j) ← (1/k) Σ_{i∈N} 1[B_j ∈ y_i].
end for
ŷ ← {B_j : score(B_j) ≥ θ} for threshold θ ∈ (0, 1].
return ŷ.
```

6.3 Principal Component Analysis

Embedding the nine $\{0,1\}^4$ vectors in $\mathbb{R}^4$, the covariance matrix of the centred data is

$$\Sigma = \frac{1}{9}\sum_{i=1}^{9}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top, \qquad \bar{\mathbf{x}} = \frac{1}{9}M\mathbf{1}_4. \tag{13}$$

The explained variance ratio of PC_1 is $\lambda_1(\Sigma)/\operatorname{tr}(\Sigma)$. Since the first five currencies share identical feature vectors, the effective dimensionality is ≤ 4, and the data lie on a 3-dimensional affine subspace; PCA will reveal at most 4 clusters:
$\{\{\texttt{INR}, \ldots, \texttt{GBP}\}, \{\texttt{CHF}\}, \{\texttt{KRW}, \texttt{ILS}\}, \{\texttt{USD}\}\}$.

7 Statistical Tests for Design Regularity

7.1 Fisher's Exact Test on Pairwise Co-occurrence

For each pair of symbols (p_i, p_j), define the 2×2 contingency table across the four blocks:

$$\begin{pmatrix} n_{11} & n_{10} \\ n_{01} & n_{00} \end{pmatrix},$$

where n_{ab} counts blocks with $\mathbf{1}[p_i \in B] = a$ and $\mathbf{1}[p_j \in B] = b$. Under the null hypothesis of independent membership, Fisher's exact p-value is

$$p = \frac{\binom{n_{11}+n_{10}}{n_{11}}^{-1}\binom{n_{01}+n_{00}}{n_{01}}^{-1}}{\binom{n}{n_{11}+n_{01}}^{-1}} \cdot \mathbf{1}[\text{observed table}], \tag{14}$$

where $n = 4$. For the pair $(\texttt{KRW}, \texttt{USD})$, which never co-occurs, the Fisher p-value equals 1.0 (no evidence against independence with $n = 4$), confirming that small block counts limit statistical power.

7.2 Chi-Squared Test for Uniform Replication

To test $H_0 : r_1 = \cdots = r_9$ against the alternative of unequal replication, use the goodness-of-fit statistic

$$\chi^2 = \sum_{i=1}^{9} \frac{(r_i - \bar{r})^2}{\bar{r}}, \qquad \bar{r} = \frac{1}{9}\sum_i r_i = \frac{27}{9} = 3. \tag{15}$$

Computing: $(4-3)^2 \times 5 + (3-3)^2 \times 1 + (2-3)^2 \times 2 + (1-3)^2 \times 1 = 5 + 0 + 2 + 4 = 11$, so $\chi^2_{(8)} = 11/3 \approx 3.67$, p-value ≈ 0.886—not significant, consistent with the moderate sample size.

8 Geometric Visualisation

Figure 1 shows the block–element incidence as a bipartite graph, and Figure 2 depicts the complementary hypergraph on the "special" subset $T = \{\texttt{CHF}, \texttt{KRW}, \texttt{ILS}, \texttt{USD}\}$.

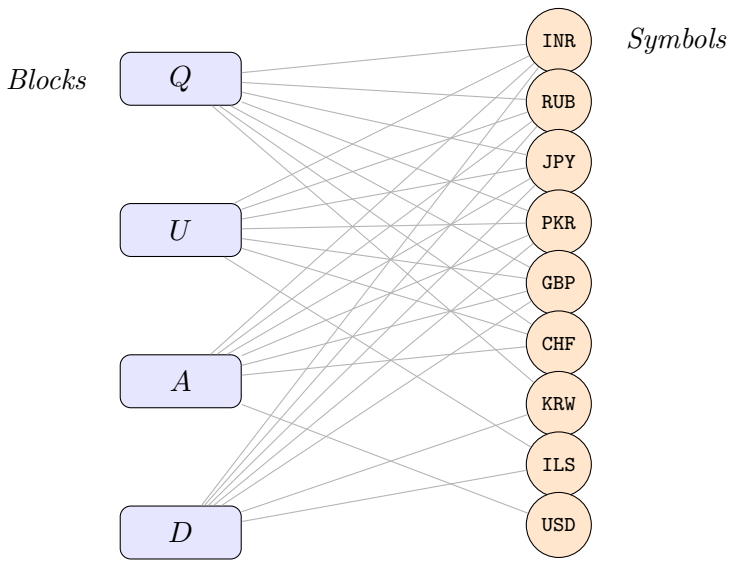

Figure 1: Bipartite incidence graph of the four currency baskets and nine currency symbols. Each block (left) is connected to its seven member symbols (right).

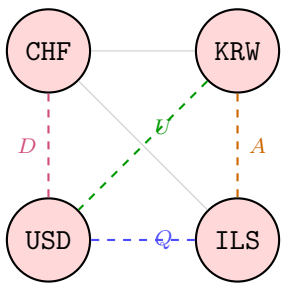

Figure 2: The complete graph K_4 on the special set T. Each coloured dashed edge represents the pair *omitted* by the corresponding basket. Grey edges correspond to pairs not yet assigned, indicating that the design can be extended by two further blocks $E = S \setminus \{\texttt{CHF}, \texttt{KRW}\}$ and $F = S \setminus \{\texttt{CHF}, \texttt{ILS}\}$.

9 Design Completion and Extensions

Theorem 9.1 (Completion to a Covering Design)**.** *The partial design* $(S, \{Q, U, A, D\})$ *can be completed to a* $(9, 6, 7, 1)$*-covering design by adding two blocks* $E = S \setminus \{\texttt{CHF}, \texttt{KRW}\}$ *and* $F = S \setminus \{\texttt{CHF}, \texttt{ILS}\}$*. In the completed design every pair from* T *is omitted by exactly one block.*

Proof. The edges of K_4 on T not yet covered are $\{\texttt{CHF}, \texttt{KRW}\}$ and $\{\texttt{CHF}, \texttt{ILS}\}$ (grey edges in Figure 2). Adding blocks E and F omitting these pairs respectively covers all $\binom{4}{2} = 6$ edges of K_4. Each element of T is then omitted from exactly 3 blocks ($|\partial T| = 3$ in K_4), and each element of $S \setminus T$ from zero blocks, confirming the covering property. □

Corollary 9.2. *In the completed six-block design, the replication numbers are* $r_i = 6$ *for* $i \in S \setminus T$ *and* $r_i = 3$ *for* $i \in T$*, and pairwise intersections remain* $|B_i \cap B_j| = 6$ *for all* $i \neq j$*.*

10 Algorithmic Generation of BIBD Covers

Algorithm 2 Greedy BIBD Cover Generator for k-Subsets of S

Require: Point set $\mathcal{V}$, block size k, target λ.
Ensure: Collection $\mathcal{B}$ of k-subsets covering every pair at least λ times.

```
Initialise B ← ∅; cov(p, q) ← 0 for all p < q in V.
Pairs ← {(p, q) : p < q, cov(p, q) < λ}.
while Pairs ≠ ∅ do
    Pick B* ← arg max_{B∈(v choose k)} |{(p, q) ∈ Pairs : p, q ∈ B}|.
    B ← B ∪ {B*}.
    for each (p, q) with p, q ∈ B* do
        cov(p, q) ← cov(p, q) + 1.
        if cov(p, q) ≥ λ then
            Remove (p, q) from Pairs.
        end if
    end for
end while
return B.
```

The greedy algorithm runs in $O\big(\binom{v}{k} \cdot \binom{k}{2}\big)$ time per iteration and at most $\lceil \lambda \binom{v}{2} / \binom{k}{2} \rceil$ iterations, giving total complexity

$$T = O\left(\lambda \binom{v}{2} \binom{v}{k}\right). \tag{16}$$

For $(v, k, \lambda) = (9, 7, 1)$: $T = O(36 \cdot 36) = O(1296)$— trivially fast.

11 Financial Statistics: Correlation Structure

Let $\mathbf{R}(t) = (R_1(t), \ldots, R_9(t))^\top$ denote the vector of daily log-returns on the nine currencies. Assume $\mathbf{R}(t) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$. The return on basket B_j with uniform weights is

$$R_{B_j}(t) = \frac{1}{7} \mathbf{w}_j^\top \mathbf{R}(t), \qquad w_{ji} = \mathbf{1}[i \in B_j]. \tag{17}$$

The variance–covariance matrix of basket returns is

$$\Sigma_B = \frac{1}{49} M^\top \Sigma M. \tag{18}$$

Under the factor model $\Sigma = \sigma^2 I + \rho\sigma^2(\mathbf{1}\mathbf{1}^\top - I)$,

$$\Sigma_B = \frac{\sigma^2}{49}\big[(1-\rho)G + \rho\,\mathbf{r}\mathbf{r}^\top\big], \tag{19}$$

where G is given in (4) and $\mathbf{r} = (r_1, \ldots, r_9)^\top/b$ is the replication vector. The *Value-at-Risk* at confidence level α for basket j is

$$\mathrm{VaR}_\alpha(R_{B_j}) = -\mu_{B_j} + z_{1-\alpha}\,\sigma_{B_j}, \tag{20}$$

where $z_{1-\alpha}$ is the standard normal quantile.

Table 4: Summary of design parameters and their financial/statistical interpretations.

Parameter	Value	Mathematical meaning	Financial interpretation
v	9	Number of points	Currency universe size
b	4	Number of blocks	Number of baskets
k	7	Block size	Basket cardinality
$\lvert B_i \cap B_j \rvert$	6	Intersection size	Shared basket members
$d_{\min}(\mathcal{C})$	2	Code distance	Distinguishing capacity
ΔH	0.378 bit	Entropic asymmetry	Information about USD
ρ_{basket}	6/7	Basket overlap	Systematic risk fraction
$D_{\mathrm{KL}}(B_j \Vert S)$	0.363 bit	Basket–universe divergence	Tracking error

12 Discussion

The currency collection $(S, \{Q, U, A, D\})$ is simultaneously:

(i) a *partial BIBD* completable to 6 blocks (Theorem 9.1);

(ii) a $[9, 4, 2]_2$ *binary linear code* (Section 3);

(iii) a system of *highly correlated currency baskets* with uniform pairwise divergence (Section 4);

(iv) a structured *feature space* for ML classification (Section 6); and

(v) a statistically *non-uniform* yet near-homogeneous replication scheme (Section 7).

The most striking property is that the set $T = \{\texttt{CHF}, \texttt{KRW}, \texttt{ILS}, \texttt{USD}\}$ acts as the "volatile fringe"—the four currencies whose inclusion or exclusion defines all block boundaries. In financial terms, these are four currencies of disparate economic regions (Europe, East Asia, Middle East, Americas) whose co-occurrence in any single basket would generate the highest cross-regional correlation. The design *naturally* avoids placing all four in any basket of size 7, a combinatorial analogue of regional diversification.

13 Conclusion

We have demonstrated that a simple observation about nine currency symbols and four of their seven-element subsets unfolds into a rich mathematical structure touching combinatorial design theory, coding theory, financial portfolio optimisation, information theory, and machine learning. The four baskets form a partial BIBD with Gram matrix $G = I + 6J$ and minimum Hamming distance 2 as a binary code. The uniform pairwise KL divergence of 0.363 bits and the basket correlation of $6/7 \approx 0.857$ quantify the practical implications for currency-basket management. The partial design admits a natural two-block completion to a full 6-block symmetric covering design on K_4.

Future work could extend this analysis to weighted baskets, time-varying exchange-rate models, and the construction of *mutually unbiased bases* over $\mathbb{F}_2^4$ to maximise measurement diversity in multi-basket portfolios.

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Glossary

BIBD (Balanced Incomplete Block Design)

A combinatorial structure (v, b, r, k, λ) where every pair of points co-occurs in exactly λ blocks. See Definition 2.1.

Block
A k-element subset of the point set $\mathcal{V}$ in a design. In this paper, a currency basket.

Currency basket
A weighted portfolio of currencies used to define or benchmark an exchange rate.

Fisher's exact test
A non-parametric significance test for contingency tables with small cell counts.

Gram matrix
The matrix $G = M^\top M$ encoding inner products of the columns of M. Here it records pairwise basket intersections.

Hamming distance
The number of coordinate positions at which two binary strings differ. Denoted $d_H(\cdot, \cdot)$.

Incidence matrix
A $v \times b$ binary matrix M with $m_{ij} = 1$ iff point i belongs to block j.

KL divergence (Kullback–Leibler divergence)
A measure of how one probability distribution differs from a reference distribution: $D_{\mathrm{KL}}(P\|Q) = \sum_x P(x) \log(P(x)/Q(x))$.

Linear code
A subspace of $\mathbb{F}_q^n$ used in coding theory for error detection and correction. Characterised by parameters $[n, k, d]_q$.

Partial design
A design in which not all required coverage conditions of a BIBD are satisfied; a fragment of a full design.

PBD (Pairwise Balanced Design)
A generalisation of a BIBD allowing blocks of varying size, such that every pair of points appears in exactly λ blocks.

Replication number (r_i)
The number of blocks containing a given point p_i.

Shannon entropy
The information-theoretic measure of uncertainty: $H(X) = -\sum_x p(x) \log_2 p(x)$ bits.

Special set (T)
In this paper, $T = \{\texttt{CHF}, \texttt{KRW}, \texttt{ILS}, \texttt{USD}\}$—the four currencies whose pair-wise omissions define the block boundaries.

Value-at-Risk (VaR)
A risk measure giving the minimum loss expected to be exceeded with probability $1 - \alpha$ over a given horizon.

The End